Short-term exposure to SO2 and all-cause and respiratory mortality: a systematic review and meta-analysis

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Review question

In any population, including subgroups of susceptible adults and children (P), what is the increase in risk (incidence/prevalence) of all-cause and respiratory mortality (O) per unit increase (C) in µg/m3 of short-term exposure (from minutes to seven days) to ambient concentration of SO2 (E), observed in studies relevant for the health outcome and exposure duration of interest (S)? In these studies, what is the lowest concentration that produces a measurable increase in risk?

Searches

Studies matching the PECOS questions will be searched comprehensively in the following bibliographic databases and citation indexes: Medical Literature Analysis and Retrieval System Online (MEDLINE) via PubMed, and Scopus via Elsevier. Additionally, regional databases (LILACS, WPRIM, IMSEAR, IMEMR, AIM) will be searched to retrieve articles not available in global databases, including thesis, conference proceedings and other grey literature.

References of identified relevant articles (and reviews/guidelines) will be scanned to identify additional published data matching the PECOS question. When the same study population is used in several publications, only the largest and the most complete study (i.e. multi-cities study) will be included. In addition, single-city study with different time periods from multi-cities study will be accepted [6]. When the same study shows different effect sizes for different countries, cities, areas, or age groups, all the effect sizes will be considered as independent from each other and will be included separately in the analysis.

Data search will include studies from January 1980 up to December 2018. An update of the data search will be performed 1 month before finalization of the systematic review in order to incorporate any relevant studies that might have been published shortly before.

A literature search strategy using free text and medical subject headings (MeSH) terms will be developed for each database, considering all the eligibility criteria derived from the PECOS questions. The search terms will be organized in three major groups: terms related to mortality, the exposure (selected pollutant), and the short-term (acute) relation between the pollutant and mortality.

Search strategy

https://www.crd.york.ac.uk/PROSPEROFILES/120738_STRATEGY_20181228.pdf

Types of study to be included

Human epidemiological studies including time-series, case-crossover and panel studies (including studies reported as cohort studies, but analysing short-term exposures). Systematic reviews of the above studies to scan for references (or as a basis for an update). Published (or accepted for publication i.e. in press) studies in peer reviewed indexed journals in any language (abstract in English language), conference abstracts and papers, letters, notes, grey literature. If suitable articles are identified published in languages not known by the SR team, further assistance will be sought after. Exclusion: qualitative studies, reviews and methodological papers, non-human studies (in vivo, in vitro, other), and studies with complete geographical and temporal overlap during meta-analysis (to avoid inclusion of patients across multiple studies).

Condition or domain being studied

All-cause and respiratory mortality.

Participants/population
General human population (including sub-groups at risk: children, pregnant women, elderly, or patients with particular conditions), of all ages, developed and developing areas, both urban and rural. No geographical restrictions. Exposure to the pollutant of interest via inhalation through ambient air predominantly. Studies on health effects of indoor exposures will not be mixed with outdoor air pollution studies in the same systematic review. If appropriate, these studies may be considered in separate reviews. Exclusion: exposure to the pollutant of interest in occupational settings.

Intervention(s), exposure(s)
Short-term exposure (from minutes to seven days) to ambient air SO2 from any source expressed in a concentration unit (µg/m3).

Comparator(s)/control
Increase in concentration unit of exposure to the air pollutant of interest in the same or in a similar population.

Context

Main outcome(s)
All-cause and respiratory mortality.

Additional outcome(s)
None.

Data extraction (selection and coding)
PO and JR will independently screen titles and abstracts identified with the systematic search and identify those that can be excluded based on the eligibility criteria. The remaining articles resulting from this selection will be assessed again by the same reviewers based on the full-text to ensure that those meet all the eligibility criteria. Any disagreement on inclusion will be resolved by discussion and, if no consensus can be reached, a third reviewer (NQ) will be consulted. Additional information from study authors (where necessary) to resolve questions about eligibility will be requested, within an acceptable time frame. Reasons for excluding articles at this stage will be recorded. Full text screening and subsequent reviewer's agreement will result in a list of included studies for systematic review.

Data extraction will be conducted in duplicate by PO and JR. If disagreement occurs, this will be resolved by discussion. A pre-designed extraction form built upon a WHO generic form and developed in Excel® will be employed.

The following characteristics of the included studies will be extracted:

- Citation details (e.g. title, authors, date of publication), type of design, study location (country/city/area).

- Characteristics of the study population.

- Details on exposure (including unit of measurement, time-lags, lag patterns, concentration range, 5th and 95th percentiles and metric description e.g. annual mean, period of year of exposure assessment, e.g. either all-year or 'warm season').

- Details on co-exposures.

- Details on outcome assessment, including International Classification of Diseases (ICD) codes for cause-specific mortality.

- Missing data management.

- Details of confounders adjusted for.

- Assessment of the shape of the concentration response function.

- Conflicts of interest.

In the absence of complete descriptions of exposure assessment and outcomes, effect estimates, or other important information, reviewers will ask authors for this information, within an acceptable time frame.

Risk of bias (quality) assessment

A risk of bias (RoB) assessment for all individual studies included in the systematic review will be conducted, and the constructs being assessed and a definition for each (domains) will be listed.

A new domain-based RoB assessment tool, developed by a group of experts convened by WHO, will be used for assessment. RoB assessment will be conducted in duplicate by two independent assessors. A WHO methods expert will support the application of the tool.

Reviewer judgment options (e.g. high, moderate, low), the number of assessors, experience of assessors (training, piloting, previous risk of bias assessment experience) will be reported.

Planned methods to summarise risk of bias assessments across studies (how the individual assessment of studies will impact the overall judgment?) and a description of how risk of bias assessments will be incorporated into data synthesis (that is, subgroup or sensitivity analyses) and their potential influence on findings will be provided by the WHO expert group.

Strategy for data synthesis

In case three or more studies are identified for the same pollutant and health outcome, a meta-analysis will be performed. We will further evaluate the shape of the exposure response function plotting the RR versus the mean concentration in a study or calculating the pooled RR for studies with an average or maximum concentration below certain values only. When multiple lag-estimates are reported by papers, the framework adopted will be: if only one lag estimate for a given pollutant/outcome pair is demonstrated (either the only one was analysed or reported), it will be included for analyses. If multiple lag estimates are reported, the selection algorithms will be: 1) the most frequently used lag in all selected studies; 2) single lags, but not cumulative/distributed lags.
RRs will be used as the common effect measure of association. If ORs are reported in the study and the outcome prevalence is lower than 10%, they will be recalculated as RRs. Meta-analyses input data may be RR for a standardised increment in pollutant concentration (10 µg/m^3), assuming a linear exposure-outcome relationship [9].
Effects expressed as interquartile (or quintile or percentile differences) should be converted into effects per concentration unit increase. Effects expressed as percentage change in mortality may also be recalculated to reflect a concentration unit increase in the pollutant by assuming a linear relationship of all pollutants.

All analyses and graphics will be performed using the “meta” package (version 4.9-2) in the statistical software R, version 3.4.4.

Statistical heterogeneity of effect estimates between studies (also inconsistency of study results) will be assessed using ?^{2} (?2), in a random effects meta-analysis. In addition, I^2 values of 30% and 50% will be take as cut-off points for mild, moderate, and severe heterogeneity.

Analysis of subgroups or subsets

Most evidence of adverse health effects would be at the population level, but subgroup analyses of the different patient populations will be performed (age group, gender, geographical area). Analysis based on multi-pollutant scenarios and on single lags will be implemented as sensitivity analyses.

Contact details for further information

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Conflicts of interest

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Subject indexing assigned by CRD

Subject index terms

Air Pollution; Humans; Sulfur Dioxide

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Details of any existing review of the same topic by the same authors

Stage of review at time of this submission

Stage	Started	Completed
Preliminary searches	Yes	Yes
Piloting of the study selection process	Yes	No
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Versions

29 January 2019

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